Practica

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Metodo de coeficientes a determinar

a) $y'' - 8y' + 16y = 3x^3 + x^2 + 2x + 6$

$y'' - 8y' + 16y = 0$

$m^2 - 8m + 16 = 0$

$(m-4)^2 = 0$

$m = 4$

$= (C_1 + C_2 x)e^{4x}$ //

$Y_p(x) = Ax^3 + Bx^2 + Cx + D$

$y'p(x) = 3Ax^2 + 2Bx + C$

$y''p(x) = 6Ax + 2B$

$(6Ax + 2B) - 8(3Ax^2 + 2Bx + C) + 16(Ax^3 + Bx^2 + Cx + D) = 3x^3 + x^2 + 2x + 6$

$6Ax + 2B - 24Ax^2 - 16Bx - 8C + 16Ax^3 + 16Bx^2 + 16Cx + 16D = 3x^3 + x^2 + 2x + 6$

$16A = 3 \Rightarrow A = \frac{3}{16}$

$-24A + 16B = 1 \rightarrow -24\left(\frac{3}{16}\right) + 16B = 1 \rightarrow -\frac{72}{16} + 16B = 1$

$16B = 1 + \frac{72}{16} = \frac{88}{16} = \frac{11}{16}$

$6A - 16B + 16C = 2$

$6\left(\frac{3}{16}\right) - 16\left(\frac{11}{16}\right) + 16C = 2$

$\frac{18}{16} - 11 + 16C = 2 \rightarrow \frac{18}{16} - \frac{176}{16} + 16C = 2$

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$$-\frac{158}{16} + 16C = 2$$

$$16C = 2 + \frac{158}{16} = \frac{190}{16}$$

$$C = \frac{95}{16}$$

$$3B - 8C + 16D = 6$$

$$2\left(\frac{11}{16}\right) - 8\left(\frac{95}{16}\right) + 16D = 6$$

$$\frac{22}{16} - \frac{760}{16} + 16D = 6 \rightarrow -\frac{738}{16} + 16D = 6$$

$$16D = 6 + \frac{738}{16} = \frac{96}{16} + \frac{738}{16} = \frac{834}{16} \rightarrow D = \frac{834}{16 \cdot 16} = \frac{417}{16}$$

$$= \frac{3}{16}x^3 + \frac{11}{16}x^2 + \frac{95}{16}x + \frac{417}{16}$$

$$= (C_1 + C_2 x)e^{4x} + \frac{3}{16}x^3 + \frac{11}{16}x^2 + \frac{95}{16}x + \frac{417}{16}$$

b) $y'' + 4y' + 3y = -e^{-x}(2 + 8x)$

$$m^2 + 4m + 3 = 0$$

$$r = \frac{-4 \pm \sqrt{4^2 - 4(1)(3)}}{2(1)} = \frac{-4 \pm \sqrt{16 - 12}}{2} = \frac{-4 \pm \sqrt{4}}{2}$$

$$r = \frac{-4 \pm 2}{2} \qquad r_1 = \frac{-4-2}{2} = -3 \qquad r_2 = \frac{-4+2}{2} = -1$$

$$= C_1 e^{-x} + C_2 e^{-3x}$$

$$y_p(x) = e^{-x}(Ax + B)$$

$$y'_p(x) = -e^{-x}(Ax + B) + e^{-x}A = e^{-x}(-Ax - B + A)$$

$$y''_p(x) = -e^{-x}(-Ax + A - B) + e^{-x}(-A) = e^{-x}(Ax - A + B - A)$$

$$y''_p(x) = e^{-x}(Ax - 2A + B)$$

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$e^{-x}(Ax-2A+B)+4e^{-x}(-Ax+A-B)+3e^{-x}(Ax+B)=-e^{-x}(2+8x)$

$e^{-x}[(Ax-2A+B)+4(-Ax+A-B)+3(Ax+B)]=-e^{-x}(2+8x)$

$(Ax-2A+B)+(-4Ax+4A-4B)+(3Ax+3B)=-2-8x$

$(A-4A+3A)x+(-2A+B+4A-4B+3B)=-8x-2$

$0x+(2A-B)=-8x-2$

$A=-8$

$2A-B=-2$

$2(-8)-B=-2 \quad -16-B=-2 \quad B=-14$

$y(x)=C_1e^{-x}+C_2e^{-3x}+e^{-x}(-8x-14)$

$=(C_1-8x-14)e^{-x}+C_2e^{-3x}$ //

d) $y''-4y'+4y=x^2e^{2x}=e^{2x}$

$m^2-4m+4=0$

$(m-2)^2=0$ // $()^2$

$((m-2)^2)^2=0^2$

$m-2=0 \rightarrow m=2$ //

$=(C_1+C_2x)e^{2x}$

$y_p(x)=e^{2x}(Ax^2+Bx+C)$

$y'p(x)=e^{2x}(2Ax^2+2Bx+2C)+e^{2x}(2Ax+B)$

$y'p(x)=e^{2x}(2Ax^2+(2B+2A)x+2C+B)$

$y''p(x)=e^{2x}(4Ax^2+4Bx+4C)+e^{2x}(2Ax^2+(2B+2A)x+2C+B)$

$=e^{2x}(4Ax^2+(4B+4A)x+4C+2Ax^2+(2B+2A)x+2C+B$

$=e^{2x}(6Ax^2+(6B+6A)x+6C+B)$

$e^{2x}(6Ax^2+(6B+6A)x+6C+B)-4e^{2x}(2Ax^2+(2B+2A)x+2C+B)+4e^{2x}$

(Ax^2+Bx+C)

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$e^{2x}[(6A-8A+4A)x^2+((6B+6A)-8(B+A)+4B)x+(6C+2B-8C-4B+4C)]$

$e^{2x}[2Ax^2+(6B-2A-4B)x+(2C-3B)]$

$e^{2x}[2Ax^2+(6B-2A-4B)x+(2C-3B)]$

$2Ax^2+(6B-2A-4B)x+(2C-3B)=x^2+1$

$2A=1 \rightarrow A=\frac{1}{2}$

$6B-2A-4B=0 \rightarrow 2B-1=0 \rightarrow B=\frac{1}{2}$

$2C-3B=-1 \rightarrow 2C-3\left(\frac{1}{2}\right)=-1 \rightarrow 2C-\frac{3}{2}=-1$

$2C=-1+\frac{3}{2} \rightarrow C=-\frac{1}{2}+\frac{3}{4}+\frac{1}{2}=\frac{1}{4}$

$y_p(x)=e^{2x}\left(\frac{1}{2}x^2+\frac{1}{2}x+\frac{1}{4}\right)$

$y(x)=(c_1+c_2x)e^{2x}+e^{2x}\left(\frac{1}{2}x^2+\frac{1}{2}x+\frac{1}{4}\right)$

$=\left(c_1+c_2x+\frac{1}{2}x^2+\frac{1}{2}x+\frac{1}{4}\right)e^{2x}$

e) $y''+y'+\frac{1}{4}y=e^x(\cos(3x)-\sin(3x))$

$m^2+m+\frac{1}{4}=0$

$r=\frac{-1\pm\sqrt{1^2-4(1)\left(\frac{1}{4}\right)}}{2(1)}$

$=\frac{-1\pm\sqrt{1-1}}{2}=\frac{-1\pm 0}{2}=-\frac{1}{2}$

$y_h(x)=(c_1+c_2x)e^{-\frac{1}{2}x}$

$y_p(x)=e^x(A\cos(3x)+B\sin(3x))$

$y'_p(x)=e^x[(A\cos(3x)+B\sin(3x))+(-3A\sin(3x)+3B\cos(3x))]$

$=e^x[(A+3B)\cos(3x)+(B-3A)\sin(3x)]$

$y''_p(x)=e^x[(A+3B)\cos(3x)+(B-3A)\sin(3x)]+e^x[(-3(A+3B)\sin(3x)+3(B-3A)\cos(3x))]$

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$(4A + 8Ax + 8B)e^{2x} - 3(2A + 4Ax + 4B)e^{2x} + 3(A + 2Ax + 2B)e^{2x} - (Ax + B)e^{2x} = 6xe^{2x}$

$[4A - 6A + 3A - A] + [8Ax - 12Ax + 6Ax - Ax] + [8B - 12B + 6B - B] = 6x$

$0A + (1Ax) + (1B) = 6x$

$1A = 6 \rightarrow A = 6$

$1B = 0 \rightarrow B = 0$

$y_p(x) = 6xe^{2x}$

$y(x) = (C_1 + C_2x + C_3x^2)e^x + 6xe^{2x}$

b) $y''' + 8y'' = -6x^2 + 9x + 2 \quad y(0) = 1 \quad y'(0) = 0$

$y''(y' + 8) = 0$

$y'' + 8y' = 0$

$m^2 + 8m = 0$

$m(m + 8) = 0$

$m_1 = 0 \quad m_2 = -8$

$= C_1 + C_2e^{-8x}$

$y_p(x) = Ax^3 + Bx^2 + Cx$

$y'_p(x) = 3Ax^2 + 2Bx + C$

$y''_p(x) = 6Ax + 2B$

$y'''_p(x) = 6A$

$6A + 8(6Ax + 2B) = -6x^2 + 9x + 2$

$6A + 48Ax + 16B = -6x^2 + 9x + 2$

$0 = -6$

$48A = 9 \rightarrow A = \frac{9}{48} = \frac{3}{16}$

$6A + 16B = 2 \rightarrow 6\left(\frac{3}{16}\right) + 16B = 2$

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$\frac{18}{16} + 16B = 2 \Rightarrow 16B = 2 - \frac{9}{8}$

$16B = \frac{7}{8} \Rightarrow B = \frac{7}{128}$

$y_p(x) = \frac{3}{16}x^3 + \frac{7}{128}x^2$

$y(x) = C_1 + C_2 e^{-8x} + \frac{3}{16}x^3 + \frac{7}{128}x^2$

Aplicamos las condiciones

$C_1 + 0 + 0 + 0 = 1$

$y'(x) = -8C_2 e^{-8x} + \frac{9}{16}x^2 + \frac{7}{64}x$

$y'(0) = -8C_2 = 0$

$y''(x) = 64C_2 e^{-8x} + \frac{9}{8}x + \frac{7}{64}$

$y''(0) = 0 + 0 + \frac{7}{64} = 0$

$y(x) = 1 + \frac{3}{16}x^3 + \frac{7}{128}x^2$

Reducción de orden

Se da una ecuación diferencial y una solución no trivial f. Determinar una segunda solución linealmente independiente

a) $x^2y'' - xy' + 2x = 0 \quad f(x) = x\sin(\ln(x))$

$y_2(x) = v(x)f(x) = v(x)\,x\sin(\ln(x))$

$y_2'(x) = v'(x)\,x\sin(\ln(x)) + v(x)[\sin(\ln(x)) + x\cos(\ln(x))]$

$y_2''(x) =$ [illegible]

$$y_p''(x) = e^x[(-8A+12B)\cos(3x) + (12A-8B)\sin(3x)]$$

$$(-8A+12B)\cos(3x) + (12A-8B)\sin(3x) + (A+3B)\cos(3x) + (B-3A)\sin(3x)$$

$$+\frac{1}{4}(A\cos(3x) + B\sin(3x)) = \cos(3x) + \sin(3x)$$

$$\left[(-8A+12B) + (A+3B) + \frac{1}{4}A\right]\cos(3x) + \left[(12A-8B) + (B-3A) + \frac{1}{4}B\right]\sin(3x)$$

$$= \cos(3x) - \sin(3x)$$

$$\begin{cases} (-8A+12B+A+3B+\frac{1}{4}A) = 1 \\ (12A-8B+B-3A+\frac{1}{4}B) = -1 \end{cases}$$

$$\begin{cases} (-7A+15B+\frac{1}{4}A) = 1 \\ (9A-7B+\frac{1}{4}B) = -1 \end{cases}$$

$$(-28A+60B+A) = 4$$

$$(36A-28B+B) = -4$$

$$(-27A+60B+A) = 4 \quad 1) \;//\; (36)$$

$$36A-27B = -4 \quad 2) \;//\; (27)$$

$$-972A+2160B = 144$$

$$972A-729B = 108$$

$$1431B = 36$$

$$B = \frac{36}{1431} = \frac{12}{477}$$

$$-27A+60\left(\frac{12}{477}\right) = 4 \rightarrow -27A+\frac{720}{477} = 4 \rightarrow -27A = 4-\frac{720}{477}$$

$$A = \frac{1}{-27}\left(\frac{4\cdot 477-720}{477}\right) = \frac{1}{-27}\left(\frac{1908-720}{477}\right) = \frac{1}{-27}\cdot\frac{1188}{477} = -\frac{44}{477}$$

$$y(x) = (C_1+C_2x)e^{\frac{x}{2}} + e^x\left(\frac{-44}{477}\cos(3x) + \frac{12}{477}\sin(3x)\right)$$

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4) $y'' + 5y' + 6y = 10e^{2x}$ $y(0) = y'(0) = 1$

$m^2 + 5m + 6 = 0$

$r = \frac{-5 \pm \sqrt{5^2 - 4 \cdot 1 \cdot 6}}{2(1)} = \frac{-5 \pm 1}{2}$

$r_1 = -2 \quad r_2 = -3$

$= C_1 e^{-2x} + C_2 e^{-3x}$

$y_p(x) = Ae^{2x}$

$y'_p(x) = 2Ae^{2x}$

$y''_p(x) = 4Ae^{2x}$

$4Ae^{2x} + 5(2Ae^{2x}) + 6(Ae^{2x}) = 10e^{2x}$

$(4A + 10A + 6A)e^{2x} = 10e^{2x}$

$20Ae^{2x} = 10e^{2x}$

$20A = 10$

$A = \frac{10}{20} = \frac{1}{2}$

$y_p(x) = \frac{1}{2}e^{2x}$

$y(x) = C_1 e^{-2x} + C_2 e^{-3x} + \frac{1}{2}e^{2x}$

5) $y''' - 3y'' + 3y' - y = 6xe^{2x}$

$m^3 - 3m^2 + 3m - 1 = 0$

$(r-1)^3 = 0$

$= (C_1 + C_2 x + C_3 x^2)e^x$

$y_p(x) = (Ax + B)e^{2x}$

$y'_p(x) = (2A + 4Ax + 4B)e^{2x}$

$y'''_p(x) = (4A + 8Ax + 8B)e^{2x}$

$y_2'' = v''(x) \cdot \sin(\ln(x)) + 2v'(x)\left(\sin(\ln(x)) + \cos(\ln(x))\right) + v(x)\left(\cos(\ln(x)) - \sin(\ln(x))\right)$

$x^2 y_2'' = x^2\left(v''(x)\sin(\ln(x)) + 2v'(x)\left(\sin(\ln(x)) + \cos(\ln(x))\right) + v(x)\left(\cos(\ln(x)) - \sin(\ln(x))\right)\right)$

$= x^3 v''(x)\sin(\ln(x)) + 2x^2 v'(x)\left(\sin(\ln(x)) + \cos(\ln(x))\right) + x^2 v(x)\left(\cos(\ln(x)) - \sin(\ln(x))\right)$

$-xy_2' = -x\left(v'(x)\sin(\ln(x)) + v(x)\left(\sin(\ln(x)) + \cos(\ln(x))\right)\right)$

$= -x^2 v'(x)\sin(\ln(x)) - xv(x)\left(\sin(\ln(x)) + \cos(\ln(x))\right)$

$2y_2 = 2v(x)\sin(\ln(x))$

$x^3 v''(x)\sin(\ln(x)) + 2x^2 v'(x)\left(\sin(\ln(x)) + \cos(\ln(x))\right) + x^2 v(x)\left(\cos(\ln(x)) - \sin(\ln(x))\right) - x^2 v'(x)\sin(\ln(x)) - xv(x)\left(\sin(\ln(x)) + \cos(\ln(x))\right) + 2v(x)\sin(\ln(x)) = 0$

$x^3 v''(x)\sin(\ln(x)) + x^2\left(v'(x)\left(\sin(\ln(x)) + 2\cos(\ln(x))\right) + v(x)\left(\cos(\ln(x)) - \sin(\ln(x))\right) - x^2 v'(x)\sin(\ln(x)) - xv(x)\left(\sin(\ln(x)) + \cos(\ln(x))\right) + 2v(x)\sin(\ln(x)) = 0$

Simplificamos

$x^3 v''(x) = 0$

$v'(x) = \frac{1}{x^2}$ // $\int$

$\int v'(x) = \int \frac{1}{x^2}$

Aplicamos regla de Potencia

$v(x) = -\frac{1}{x} + C$

$y_2(x) = v(x)f(x) = -\frac{C}{x} \cdot x\sin(\ln(x)) = -C\sin(\ln(x))$

$y(x) = C_1 x\sin(\ln(x)) + C_2\sin(\ln(x))$

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b) $y'' - 25y = 0 \quad f(x) = e^{5x}$

$$y_2(x) = v(x)e^{5x}$$

$$y_2'(x) = v'(x)e^{5x} + 5v(x)e^{5x}$$

$$y_2''(x) = v''(x)e^{5x} + 10v'(x)e^{5x} + 25v(x)e^{5x}$$

$$v''(x)e^{5x} + 10v'(x)e^{5x} + 25v(x)e^{5x} - 25(v(x)e^{5x}) = 0$$

$$v''(x)e^{5x} + 10v'(x)e^{5x} = 0$$

$$v''(x) + 10v'(x) = 0$$

$$v''(x) = -10v'(x)$$

$$v'(x) = C_1 e^{-10x}$$

$$v(x) = \int C_1 e^{-10x}\,dx = -\frac{C_1}{10}e^{-10x} + C_2$$

$$y_2(x) = \left(-\frac{C_1}{10}e^{-10x} + C_2\right)e^{5x}$$

$$y_2(x) = -\frac{C_1}{10}e^{-5x} + C_2e^{5x}$$

$$y(x) = C_1e^{5x} + C_2e^{-5x}$$

c) $x y'' + (1-2x)y' + (x-1)y = 0 \quad f(x) = e^x$ para $x \to 0$

$$y_2(x) = v(x)e^x$$

$$y_2'(x) = v'(x)e^x + v(x)e^x$$

$$y_2'(x) = e^x(v'(x) + v(x))$$

$$y_2''(x) = e^x(v''(x) + v'(x)) + e^x(v'(x) + v(x))$$

$$y_2''(x) = e^x(v''(x) + 2v'(x) + v(x))$$

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$$x\left[e^x v''(x) + 2v'(x) + v(x)\right] + (1-2x)\left[e^x(v'(x) + v(x)\right] + (x-1)\left[e^x v(x)\right] = 0$$

$$e^x\left[x(v''(x) + 2v'(x) + v(x)) + (1-2x)(v'(x) + v(x)) + (x-1)v(x)\right] = 0$$

$$x(v''(x) + 2v'(x) + v(x)) + (1-2x)(v'(x) + v(x)) + (x-1)v(x) = 0$$

$$xv''(x) + 2xv'(x) + xv(x) + (1-2x)v'(x) + (1-2x)v(x) + (x-1)v(x) = 0$$

$$xv''(x) + v'(x) = 0$$

$$v''(x) + \frac{1}{x}v'(x) = 0$$

$$w(x) = v'(x)$$

$$w'(x) + \frac{1}{x}w(x) = 0$$

$$w(x) = \frac{C_1}{x}$$

$$v(x) = \int \frac{C_1}{x}\,dx = C_1 \ln(x) + C_2$$

$$y_2(x) = (C_1 \ln(x) + C_2)e^x$$

$$Y_2(x) = \ln(x)e^x$$

$$Y(x) = C_1 e^x + C_2 \ln(x) e^x$$

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